# DevOps Viva & Folder Presentation Guide

## Section 1: Explaining the Directory Structure

When representing the project files to your examiner (Mam), explain them as a continuous, automated flow:

- **app/ :** Contains index.html, style.css, script.js representing the front-end application.
- **Dockerfile & nginx.conf :** Houses Nginx configuration and Docker build file to containerize the web app.
- **terraform/ :** Infrastructure as Code (IaC) configuration setting up VPC, Security Group, EC2 instance, and Elastic IP.
- **Jenkinsfile :** Jenkins automated CI/CD pipeline script containing checkout, build, push, deploy stages.
- **.github/workflows/ :** Modern GitOps automation workflow acting as secondary continuous integration tool.
- **monitoring/ :** Configures Prometheus to scrape and observe server metrics from the host.

## Section 2: Key Viva Questions & Answers

### Docker & Containerization

**Q: Why did you containerize your application? Why use Docker?**

Answer: Docker ensures environment consistency. It packages the application code together with dependencies (Nginx server) so it runs identically in development, testing, and production without dependency drift.

**Q: What is the base image used in your Dockerfile?**

Answer: We used nginx:alpine. Alpine is chosen because it is an extremely small, secure, and lightweight Linux distribution, making the final container image fast to pull and secure against vulnerabilities.

## Terraform (Infrastructure as Code)

### Q: What is the advantage of using Terraform over the AWS Web Console?

Answer: Terraform allows us to write Infrastructure as Code (IaC). This makes server setups repeatable, trackable in Git, and fully automated. We can deploy/destroy environments in seconds without making manual errors.

### Q: What is the role of the terraform.tfstate file?

Answer: The state file maps real-world AWS resources to your Terraform configuration. It allows Terraform to know which resources already exist so it only creates or modifies what is necessary.

## CI/CD & Monitoring

### Q: Explain the stages in your Jenkins pipeline.

Answer: The pipeline contains 5 major stages: Checkout (fetches source code), Build (builds Docker image), Push (pushes to Docker Hub), Terraform Apply (provisions AWS infrastructure), and Deploy (SSH into EC2 to pull and run the new container).

### Q: What is the role of Prometheus in your architecture?

Answer: Prometheus is a time-series database and monitoring tool. It collects metrics from the server (CPU load, RAM usage, request counts) via HTTP scraping, allowing us to monitor system health and set up alerts.